ShwaasAI – Repo Analysis & Full-Stack Work Plan

INNOTECH 2026 · Team KARTAVYA · Analysis of the `Shwaas-AI-main` repo and the project deck, prepared for the full-stack, web and app owner.

The repo contains the ML code and a FastAPI backend only. There is no app, no web code and no hardware code, so the app and web side starts from zero.

What's implemented

Area	Status
FastAPI `POST /predict` with the full pipeline	Done in `src/api.py` (276 lines)
Pipeline contents	Age gate, quality gate, bandpass filter, cough extraction, mel-spectrogram, MobileNetV2, temperature scaling, clinical scoring, tier routing, optional Grad-CAM
Training, baselines, evaluation, plots	Done, with metrics JSONs and 8 report images
Model weights	Pulled from Hugging Face (`Kushagra-01/shwaasai-model`) at startup
Mobile app, web dashboard, DB, auth, sessions	**Not in the repo**
Hardware code (ESP32)	**Not in the repo**

Overall, about 40% is done: the backend is around 80%, and the app, web, persistence and hardware are around 0%.

Backend problems that will hit the app

These need to be fixed with Kushagra before the app is built against the API.

1. **The README doesn't match the code.**

- The README documents `q1`–`q4` plus a `verification.status == "clarification_required"` flow.
- The real endpoint takes 10 named fields (`cough_duration`, `fever`, `night_sweats`, and so on) plus `age` and `sex`, and has no clarification logic at all.
- It also has no session ID.
- The deck matches the code, not the README.

2. **The answer encoding is inconsistent.**
   - The docs say `0 = No, 1 = Yes, 2 = Unsure`.
   - But `calculate_clinical_score` simply sums the values, so "Unsure" scores higher than "Yes".
   - The red flag fires on `hemoptysis == 2`, which the docs define as "Unsure".
   - `tb_contact == 3` can never be true.
   - Ten fields at a maximum of 2 each gives 20, not the "0–18" in the deck.
   - Ask Kushagra what each answer value means before building the question UI.
3. **No CORS middleware.** A browser-based dashboard will fail until `CORSMiddleware` is added.
4. **Blocking and thread-safety issues.**
   - `/predict` is `async` but runs CPU-heavy work, so it blocks the event loop.
   - The Grad-CAM hooks store state globally, so concurrent requests can corrupt each other's heatmaps.
   - It's fine for a demo with one user, but not under load.
5. **The response has no stable enums.** `recommended_pathway` is a free-text sentence, so the app shouldn't parse it. Ask for a machine-readable `pathway` code alongside it.
6. **Weights download at import.** The first startup needs internet access. Bake the weights into the Docker image if deploying.

Problems in the ML claims (flag to Kushagra)

- `train.py` trains the fusion head on **synthetic tabular data correlated with the label** (a comment in the code calls it a "PROXY TASK HACK"). `reports/cnn_honest_metrics.json` shows the resulting 0.954 AUROC, which is inflated. The 0.791 figure is with the tabular input zeroed, and that is what the API serves, so train and serve inputs differ.

- The confusion matrix in the deck sums to 637 samples, but the metrics JSON says 325 test participants. Make sure the slide's numbers come from a documented run.
- The README itself says the 0.02 threshold is only valid if it was chosen on validation data. If it was picked on the test set, a judge could challenge it.

Work plan, in priority order

1. Define the API contract (half a day)

Agree with Kushagra on the answer encoding, a `session_id`, a stable `tier` enum, and a `pathway` code. Add `GET /health` and CORS.

2. Mobile app (the core deliverable)

- Consent screen and intake form with age, sex and the 10 three-option questions.
- 3 s recorder at 16 kHz mono WAV, with a countdown and level meter, and a toggle to use the phone mic until the ESP32 is ready.
- Multipart upload with loading, timeout and retry states.
- Error screens: a 400 with `route_to_pediatric_protocol` shows the pediatric screen, and a 400 with `request_new_recording` shows a retry screen. Both are normal flows, not crashes.
- Result screen with a tier badge, the recommended action, scores, optional Grad-CAM from the base64 data URI, and the line "screening support, not a diagnosis".
- Offline queue.
- Flutter is a good fit here.

3. Web dashboard (the differentiator)

- The problem slide says 44% of presumptive cases are lost to follow-up, and a referral tracker addresses that directly.
- Add a small DB (SQLite or Postgres) behind FastAPI and a login.
- The dashboard needs a screenings list filtered by tier and a referral status (pending, tested, lost).
- Store the questionnaire answers and results, not the audio, since the backend deletes the audio.

4. Deployment and demo safety

Dockerize with weights baked in, give it a public URL with HTTPS, and keep sample WAVs and a canned-response mode for when venue Wi-Fi fails.

5. Hardware link

Decide how the ESP32 delivers audio to the phone (BLE, USB or Wi-Fi). If that isn't ready, the phone mic is the fallback.

Deck issues to fix before presenting

- The problem slide says 27% of the global TB burden in the left panel and 25% in the right panel.
- The confusion matrix uses threshold 0.02, but the routing logic uses 0.45 / 0.65 / 0.85. Have a one-line answer for how these relate.
- The ablation labels the model "Late-Fusion CNN (Audio Only)", which is contradictory.
- Verify the Dr. Rakesh Kumar (AIIMS) quote and the 75–80% sensitivity claim, or remove them.
- Keep the wording "respiratory abnormality proxy", never "TB detected", in the app copy.